WORKING TITLE

by

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Abstract

My thesis is about bla bla

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Thanks!

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Chapter 2

Background

2.1 Multicategories

EXPLAIN THE SOURCE OF THIS MATERIAL. "Multicategories were first defined in ..." "Our exposition follows ..." yes Multicategories are a generalization of categories. In categories, arrows have a single source/domain and a single target/codomain. In multicategories, the arrows have a finite sequence of sources/inputs and a single target/output.

Definition 2.1.1. A *multicategory* $\mathcal{M}$ consist of a set S of objects and for each $n \geq 0$ and each sequence $s_1, \ldots, s_n, t$ of objects a set

$$\mathcal{M}(s_1, \ldots, s_n; t)$$

of multiarrows/operations, thought of as taking n inputs objects $s_1, \ldots, s_n$ respectively to the output object t. Moreover, there are three kinds of structure maps:

- For each object s, an identity/unit $1_s \in \mathcal{M}(s; s)$

- For any sequence of objects $s_1, \dots, s_n, t$ and any n-tuple of sequences $d_1^i, \dots, d_{k_i}^i$ of objects for $i = 1, \dots, n$, a composition function of multiarrows/operations

  $$\gamma : \mathcal{M}(s_1, \dots, s_n; t) \times \prod_{i=1}^{n} \mathcal{M}(d_1^i, \dots, d_{k_i}^i; s_i) \to \mathcal{M}(d_1^1, \dots, d_{k_1}^1, d_1^2, \dots, d_{k_n}^n; t),$$

  usually written $p \circ (q_1, \dots, q_n) := \gamma(p, q_1, \dots, q_n)$ or even just $p(q_1, \dots, q_n)$

Furthermore these structure maps have to satisfy a number of axioms, which express that composition is unital and associative (Heuts and Moerdijk, 2022)

Definition 2.1.2. A (colored) operad $\boldsymbol{P}$ is a multicategory with a symmetric action on the set of multiarrows. That is:

- For each permutation $\sigma \in \Sigma_n$, a map

  $$\sigma^* : \boldsymbol{P}(c_1, \dots, c_n; c) \to \boldsymbol{P}(c_{\sigma(1)}, \dots, c_{\sigma(n)}; c)$$

  usually written $\sigma^* p = p \circ \sigma$

such that these maps are compatible with the composition and the units giving a right action of the symmetric groups on the sets of multiarrows.

We will be calling multicategories when we are not interested in the symmetric action, and operads when we are.

Remark 2.1.3. Let $\mathcal{M}$ be a multicategory. Given multiarrows $f \in \mathcal{M}(s_1, \dots, s_n; t)$ and $g \in \mathcal{M}(d_1, \dots, d_l; s_i)$ for some $1 \leq i \leq n$. It will be convenient to have the following notation:

$$f \circ_i g := f(1_{s_1}, \dots, g, \dots, 1_{s_n})$$

Thus, $f \circ_i g$ is the composition of f with g in the ith variable.

Example 2.1.4. You can think of n-ary multiarrows in $\mathcal{M}(s_1, \ldots, s_n; t)$ as n-ary operations. Let M be the multicategory with a single object X. Then the n-ary multiarrows in $M(X, \ldots, X; X)$define to be the set of functions

$$f : X \times \cdots \times X \to X$$

where the domain is the n-fold cartesian product of X with itself.

Example 2.1.5. We can up the level from the previous example, by fixing a family of sets $\{X_c\}_{c \in C}$ indexed by a set of objects C. Then you can think of the multiarrows in $\mathcal{M}(c_1, \ldots, c_n; c)$ as n-ary operations, taking inputs of types $c_1, \ldots, c_n$ respectively to an output of type c.

For example, for a fixed family of sets $\{X_c\}_{c \in C}$, define a multicategory $\mathcal{M}$ whose objects is the set of colors C, and whose multiarrows are defined by declaring $\mathcal{M}(c_1, \ldots, c_n; c)$ to be the set of functions

$$p : X_{c_1} \times \cdots \times X_{c_n} \to X_c$$

Composition in $\mathcal{M}$ is defined as composition of functions.

Example 2.1.6. The previous two examples are specific instances of a more general family of examples example. If $\mathcal{C}$ is a symmetric monoidal category and C is a set of objects of $\mathcal{C}$, then one obtains a multicategory with this set of objects by defining $\mathcal{M}(c_1, \ldots, c_m; t)$ to be the set of morphisms $c_1 \otimes \cdots \otimes c_n \to t$ in $\mathcal{C}$. We will sometimes write $\mathcal{C}^{\otimes}$ for the multicategory obtained in this way, when C is the set of all objects of $\mathcal{C}$ (granted it is small).

Example 2.1.7. A (small) category $\mathcal{C}$ can be considered as a multicategory $\mathcal{C}_m$ which only have unary arrows. I.e. the set $\mathcal{C}_m(c_1, \ldots, c_n; c)$ is always empty unless $n = 1$.

We can observe that multicategories form a category $\mathcal{MC}\mathsf{at}$.

Definition 2.1.8. For two multicategories $\mathcal{M}$ and $\mathcal{N}$, a *multifunctor* $\phi : \mathcal{M} \to \mathcal{N}$ consist of:

- A function $f : O_{\mathcal{M}} \to O_{\mathcal{N}}$ between the corresponding sets of objects
- For each sequence $s_1, \ldots, s_n, t$ of objects of $\mathcal{M}$, a map

$$\phi_{s_1,\ldots,s_n,t} : \mathcal{M}(s_1, \ldots, s_n; t) \to \mathcal{N}(f(s_1), \ldots, f(s_n); f(t)).$$

So that these maps are compatible with compositions and units in the obvious way.

2.1.1 The Tensor Product of Operads

Let M and N be operads. Their Boardman-Vogt tensor product $M \otimes_{BV} N$ is a operad defined to make sense of what a M-algebra in the category of N-algebras is and vice versa. We will not talk about the algebras of a multicategories here, but we refer the reader to [look for reference]

Definition 2.1.9. We define the *Boardman-Vogt tensor* $M \otimes_{BV} N$ in terms of generators and relations. The set of objects of this multicategory is the product of the sets of objects of M and N; if x and y are objects of M and N respectively, we write $x \otimes y$ for the corresponding object of $M \otimes_{BV} N$. The generating multiarrows of $M \otimes_{BV} N$ are of two types:

- For each multiarrow $f \in M(x_1, \ldots, x_n, x)$ and object y of N, a multiarrow

$$f \otimes y \in (M \otimes_{BV} N)(x_1 \otimes y, \ldots, x_n \otimes y; x \otimes y)$$

- For each object x of M and multiarrow $g \in N(y_1, \ldots, y_m; y)$, a multiarrow

$$x \otimes g \in (M \otimes_{BV} N)(x \otimes y_1, \ldots, x \otimes y_m; x \otimes y)$$

The relations to be satisfied by these generators are the following:

(1) $(f \otimes y)(f_1 \otimes y, \ldots, f_n \otimes y) = f(f_1, \ldots, f_n) \otimes y$

(2) $(x \otimes g)(x \otimes g_1, \ldots, x \otimes g_m) = x \otimes g(g_1, \ldots, g_m)$

(3) $\sigma^*(f \otimes y) = (\sigma^* f) \otimes y$ for $\sigma \in \Sigma_n$

(4) $\sigma^*(x \otimes g) = x \otimes (\sigma^* g)$ for $\sigma \in \Sigma_m$

(5) $(f \otimes y)(x_1 \otimes g, \ldots, x_n \otimes g) = \sigma^*_{n,m}((x \otimes g)(f \otimes y_1, \ldots, f \otimes y_m))$

Note that relations (1) and (2) say precisely the condition that for any object y of N there is a multifunction $- \otimes y : M \to M \otimes_{BV} N$. Similarly, for any object x of M there is a multifunction $x \otimes - : N \to M \otimes_{BV} N$, this time given by relations (2) and (4). Relation (5) is the most interesting; most often referred to it as the *Boardman-Vogt interchange relation*. The permutation $\sigma_{n,m} \in \Sigma_{nm}$ is the one that makes sense of formula (5), i.e. the element relating the sequences $(x_1 \otimes y_1, \ldots, x_1 \otimes y_m, \ldots, x_n \otimes y_1, \ldots, x_n \otimes y_m)$ and $(x_1 \otimes d_1, \ldots, x_n \otimes d_1, \ldots, x_1 \otimes d_m, \ldots, x_n \otimes d_m)$.

2.2 Trees

We will do a quick detour to talk about our main combinatorial object that will help us with studying multicategories: trees. We are mainly interested in finite rooted trees. So we don't get confused with the more general notion of tree in graph theory, we will give a precise definition of what we mean by a tree in this thesis. We will see that these trees give rise to free multicategories whose objects are the edges of the tree and morphisms are generated by the vertices of the tree.

Definition 2.2.1. A *tree* T consist of a finite set V of *vertices*, a nonempty finite set E of *edges*, a distinguished element $r \in E$ called the *root*, together with the following data:

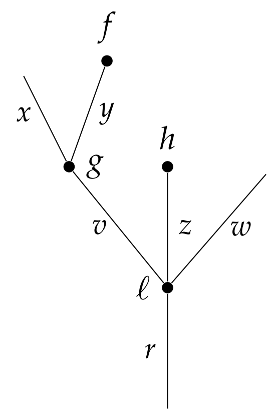
(1) A function $I : E - \{r\} \to V$, which we think of as assigning to an edge e the vertex $I(e)$ of which is an input. The way to think about this function is that "$I(e) = f$ if and only if e is *an* input edge of f"

(2) An injective function $O : V \to E$, assigning to each vertex g its output edge $O(g)$. The way to think about this function is that "$O(f) = x$ if and only if x is *the* output edge of f"

For each edge e other than the root r, we obtain a sequence of edges starting at e by repeatedly applying $O \circ I$. We demand that this sequence ends in the root r after finitely many steps, for an arbitrary starting edge.

As we can see the main difference between our definition of a tree and the usual definition of a tree in graph theory is that we allow for "orphan"/"leaf" edges that are not attached to any vertex as an output edge, and a unique designed edge that will not be attached to any vertex as an input edge (the root). This special edge also gives us a natural orientation of the tree, where we think of the root as the bottom and the leaves as the top, and thus a partial ordering on both the set of edges and the set of vertices.

Example 2.2.2. A typical tree T with $E(T) = \{x, y, z, w, v, r\}$ and $V(T) = \{f, g, h, l\}$ could look like:



We can fully define both functions for this small example as:

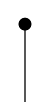
$$I(x) = I(y) = g, \;\; I(v) = I(z) = I(w) = \ell$$

$$O(f) = y, \;\; O(g) = v, \;\; O(h) = z, \;\; O(\ell) = r$$

Now I will introduce some terminology to talk about the different parts of a tree. The edges in the complement of the image of O are called *leaves* of the tree T. The vertices in the complement of the image of I are called *stumps*, or *nullary vertices.* An *outer edge* is an edge that is either the root or a leaf. An *inner edge* is any other edge of T. Such an edge is both an output edge and an input edge for some other vertex. The vertices come in two kinds, *external vertices* which only have one inner edge attached to them, and *internal vertices* When writing T for a tree, we will usually write $E(T)$ and $V(T)$ for its sets of edges and vertices, respectively.

Example 2.2.3. The smallest possible tree consists of a single edge and no vertices; this tree will be denoted by μ

Example 2.2.4. The next smallest tree consist of a single edge and single vertex, pictured as:



The valence of a vertex v is the number of input edges it has, i.e. the cardinality of the set $I^{-1}(v)$. A vertex of valence 0 is what we previously called a stump. We previously mentioned that the sets $E(T)$ and $V(T)$ both have a natural partial ordering. Expounding it, it means that $x < y$ for edges x and y (or $f < g$ for vertices f and g) if there is a sequence of edges starting at x (resp. f) and ending at the root of T, r, and passes through y (resp. g). We will call two edges x and y *incomparable* if they are not related in this partial order.

We note that removing an external vertex and all the outer edges attached to it yields a new tree. We will refer to this procedura as *pruning*.

Definition 2.2.5. A tree is called *open* if it has no nullary vertices.

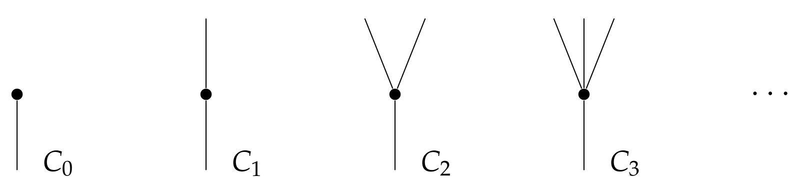
Definition 2.2.6. A tree is called *closed* if it has no leaves.

Definition 2.2.7. A tree is called *linear* if all its vertices have valence one.

As an example, the tree μ is open but not closed, and generally a tree cannot be both open and closed. A linear tree with $n+1$ edges running from the leaf 0 to the root n will be denoted by $[n]$:

The reason for this notation is that the linear trees will become clear in the following section. Note that by our definition the tree obtained from $[n]$ by putting a stump on top is not a linear tree since the stump has valence zero..

Another family of trees that we will become familiar with are the *corollas*, which are trees with a single vertex. The corolla with n leaves labeled $1, \ldots n$ and root 0 will be denoted by C_n:



Definition 2.2.8. An *embedding* $\psi : S \to T$ of trees consists of injective maps $\psi_V : V(S) \to V(T)$ and $\psi_E : E(S) \to E(T)$ such that for any vertex f of S, the map ψ_E gives a bijection between $O^{-1}(f)$ and $O_{-1}(\psi_V(f))$, and $\psi_E(O(f)) =$

$O(\psi_V(f))$. If an embedding is bijective on both edges and vertices, then it is an isomorphism of trees.

If an embedding of trees $\iota : S \to T$ is given by inclusion maps of $E(S) \subseteq E(T)$ and $V(S) \subseteq V(T)$, we will say that S is a *subtree* of T. Then we can see that any embedding $\psi : S \to T$ factors as an isomorphism followed by the inclusion of a subtree.

Definition 2.2.9. A tree T defines an operad $\mathbf{\Omega}(T)$ as follows:

- The objects of $\mathbf{\Omega}(T)$ are the edges of T.
- $\mathbf{\Omega}(T)(e_1, \ldots, e_n; e)$ is either empty or a singleton if there exist a subtree C_n with leaves $e_1, \ldots, e_n$ and root e. This set is generated by the vertices of the tree and freely composing them to get the other subtrees' operations.
- Composition is given by the tree structure in the obvious way. Expanding the details of it, we get the composition operation/multiarrow by identifying the root of the precomposing operation with a leave of the postcomposing operation and squashing this new inner edge and identifying the two vertices to the composition of the two corresponding operations/multiarrows.
- The action of Σ_n on n-ary operation is given by the relabeling of the input edges of a vertex that defines a generating operation. The other subtrees' operation follow, making sure we respect composition ordering of inputs.

2.3 Category of trees $\mathbf{\Omega}$

Dendroidal sets are a generalization of simplicial sets that are particularly well suited to study operads/multicategories and multicategories. Just as simplicial sets are presheaves on the simplex category $\mathbf{\Delta}$, dendroidal sets are presheaves on the dendroidal category $\mathbf{\Omega}$.

Definition 2.3.1. The category $\mathbf{\Omega}$ consist of:

1. Objects: finite rooted trees

2. Morphisms: for two trees S and T, a morphism from S to T is a multi-functor $\mathbf{\Omega}(S) \to \mathbf{\Omega}(T)$ between the corresponding Operads.

Another way to say this is that $\mathbf{\Omega}$ is the full subcategory of Op, the category of operads, spanned by the multicategories of the form $\mathbf{\Omega}(T)$ for some tree T.

The linear trees $[n]$ define a full subcategory of $\mathbf{\Omega}$ isomorphic to the simplex category $\mathbf{\Delta}$. Thus, there is an inclusion functor $i : \mathbf{\Delta} \to \mathbf{\Omega}$. This inclusion is compatible with the inclusion of Cat into $\mathcal{MC}\mathsf{at}$ in the sense that the following diagram commutes:

$$\begin{tikzcd}
\mathbf{\Delta} \arrow[r] \arrow[d, "i"'] & \mathsf{Cat} \arrow[d, "j"] \\
\mathbf{\Omega} \arrow[r] & \mathsf{Op}
\end{tikzcd}$$

2.4 Faces and Degeneracies in $\mathbf{\Omega}$

Recall that, the simplex category $\mathbf{\Delta}$ has two important families of morphisms, the face maps and the degeneracy maps. These maps generate all other morphisms in $\mathbf{\Delta}$. The good news for $\mathbf{\Omega}$ being some multicategorical extension of $\mathbf{\Delta}$ is that it also has two families of morphisms that generate all other morphisms in $\mathbf{\Omega}$. Like for the simplex category, the face maps in Ω have two kinds:

- The outer face maps
- The inner face maps

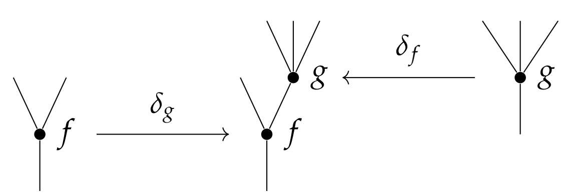
The other calls of maps are again called degeneracies.

Definition 2.4.1 (Outer Faces)**.** For a tree T and an external vertex f of T, write $\partial_f T$ for the tree obtained by pruning away f and all the outer edges attached to it. This vertex could be any leaf vertex or it can be the vertex attached to the root of T, provided it is only attached to one inner vertex of T. We write

$$\partial_f T \xrightarrow{\delta_f} T$$

for the inclusion of this tree and refer to it as an *outer face*. If f is a leaf vertex we will sometimes specifically speak of a *leaf face*, and similarly for a *root* face.

Example 2.4.2. In the tree below both vertices f and g are external vertices. The left-hand map depicts the face δ_g which is a leaf face. While the right hand map depicts the face δ_f which is an example of a root face.



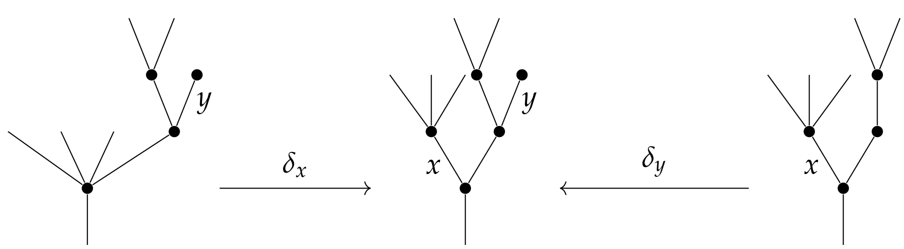
Remark 2.4.3. Corollas, by our definition, are trees with only one unique vertex and n leaves. This vertex in a corolla is not an external vertex because it is not attached to any interior edges. Nevertheless by convention, C_n has $n+1$ different maps $\mu \to C_n$ corresponding to the inclusion of each of the n leaves and the root. We define all of these to be outer faces.

Definition 2.4.4 (Inner Faces)**.** For a tree T and an inner edge x, let $\partial_x T$ denote the face of T obtained by contracting the edge x, and identifying the vertices f and g at either end of x. This is an *inner face* of T. The associated inner face map denoted

$$\partial_x T \xrightarrow{\delta_x} T$$

Should be understood not in terms of the combinatorics of trees but in terms of their associated operads. In this framework, we see that we send the new vertex in $\partial_x T$ to the composition in $\mathbf{\Omega}(T)$ of the operations correspondingly given by f and g along x. Thus the image of the new vertex in $\partial_x T$ can be identified with the subtree of T generated by the edge x. A morphism of these kind will be referred to as an *inner face*.

Example 2.4.5. The inner face $\partial_x T$ and $\partial_y T$ given by the edges x and y of the following tree T:



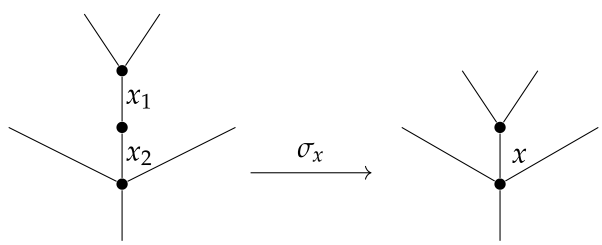
Definition 2.4.6 (Degeneracies)**.** Let x be an edge in a tree T. Consider the tree $\sigma_e T$ obtained from T by splitting this edge into two edges x_1 and x_2 by

placing an unary vertex id_x in the middle of x. The there is a morphism

$$\sigma_x T \xrightarrow{\sigma_e} T$$

in $\mathbf{\Omega}$ which sends both edges x_1 and x_2 to x and the new unary vertex id_x to the subtree $\mu \xrightarrow{x} T$ of T. In the operadic framework, we see that it sends the new vertex id_x to the identity operation of the color x in $\mathbf{\Omega}(T)$. A morphism of this kind is called a *degeneracy*.

Example 2.4.7. The degeneracy $\sigma_x T$ given by the edge x of the following tree T has the form:



Remark 2.4.8. In the special case of linear trees, the outer face maps, inner face maps, and degeneracies recover exactly the corresponding maps in $\mathbf{\Delta}$

As with simplices, we have codendroidal identities. We won't go over them since we won't be using them in any kind of way. To see a full list and more on them please refer to Heuts and Moerdijk, 2022, §3.3.4.

Remark 2.4.9. Extending the result from simplices, we can also prove that any morphism of trees can be factorized into a composition of a degeneracy, followed by an isomorphism and ending with a face map.

2.5 Dendroidal Sets

As no surprise to the reader, the definition of a *dendroidal set* is analogous to the definition of a simplicial set.

Definition 2.5.1. A dendroidal set is a functor

$$X \;:\; \mathbf{\Omega}^{\mathrm{op}} \to \mathcal{S}\mathsf{et}$$

Dendroidal sets form the objects of a category $\mathsf{d}\mathcal{S}\mathsf{et}$ whose morphisms are the natural transformations between them.

Expounding out the definition of a dendroidal sets, we can see a dendroidal set is given by a family of sets X_T, one for each $T \in \mathbf{\Omega}$, together with a map

$$\alpha^* \;:\; X_T \to X_S$$

for every morphism $\alpha : S \to T$ in $\mathbf{\Omega}$. These maps should be functorial. The elements of X_T are called *dendrices* of X of shape T, or simply T-dendrices of X.

In particular, each set X_T carries an action of the group $\mathrm{Aut}(T)$ and the dendroidal set X is completely determined by the collection of all these $\mathrm{Aut}(T)$-sets together with the elementary face and degeneracy operators of the three kinds we saw before, satisfying the *dendroidal identities*. dual to the identities previously mentioned.

Example 2.5.2. Any tree T of $\mathbf{\Omega}$ gives rise to a representable presheaf which we denote by $\mathbf{\Omega}[T]$. As in any presheaf category this are defined by

$$\mathbf{\Omega}[T]_S = \mathrm{Hom}_{\mathbf{\Omega}}(S, T)$$

Example 2.5.3. One can form limits and colimits of diagrams of dendroidal sets by calculating them 'pointwise', as in any presheaf category. For instance, we can form the product $X \times Y$ of dendroidal sets X and Y as

$$(X \times Y)_T = X_T \times Y_T$$

Example 2.5.4. In particular, each face $\delta_x : \partial_x T \to T$, where x is an inner edge or outer vertex, defines a monomorphism of dendroidal sets

$$\mathbf{\Omega}[\partial_x T] \to \mathbf{\Omega}[T].$$

We will write $\partial_x \mathbf{\Omega}[T] \subseteq \mathbf{\Omega}[T]$ for the image of this morphism and call it the *face of T opposite x*. The union of all these (inner and outer faces) defines the *boundary* $\partial \mathbf{\Omega}[T]$ of $\mathbf{\Omega}[T]$. Note that these faces and the boundary can be explicitly described as follows:

$$(\partial_x \mathbf{\Omega}[T])_S = \{\alpha : S \to T \mid \text{the edge } x \text{ is not contained in the image of } \alpha\},$$

$$(\partial_f \mathbf{\Omega}[T])_S = \{\alpha : S \to T \mid \text{no subtree } \alpha(g) \text{ contains the vertex } f\},$$

$$(\partial \mathbf{\Omega}[T])_S = \{\alpha : S \to T \mid \alpha \text{ factors through a proper inclusion } R \to T\}.$$

Example 2.5.5. The category $\mathbf{\Omega}$ has been defined as a full subcategory of the category in Op. Thus any such operad $\boldsymbol{M}$ defines a dendroidal set $\mathrm{N}\boldsymbol{M}$, its *dendroidal nerve*, by

$$\mathrm{N}\boldsymbol{M}_T = \mathsf{Op}(\mathbf{\Omega}(T), \boldsymbol{M}).$$

Even though this is an standard construction in homotopical category theory, let us take a closer look at the dendroidal set given by the operad. For the tree μ, the set $\mathrm{N}\boldsymbol{M}_\mu$ is the set of colors of $\boldsymbol{M}$. For the n-corolla C_n and a sequence of colors $(c_1, \ldots, c_n; c)$ a sequence of colors of $\boldsymbol{M}$, there is a pullback square

$$\begin{tikzcd}
\boldsymbol{M}(c_1, \ldots, c_n; c) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{N}\boldsymbol{M}_{C_n} \arrow[d] \\
* \arrow[r] & \prod_{i=0}^{n} \mathrm{N}\boldsymbol{M}_\mu
\end{tikzcd}$$

Here the right vertical map has as its i^{th} component the map $i^* : \mathrm{N}\boldsymbol{M}_{C_n} \to \mathrm{N}\boldsymbol{M}_\mu$, induced by the edge inclusion $i : \mu \to C_n$, while $\boldsymbol{c}$ denotes the sequence $(c_1, \ldots, c_n; c)$. More generally, for a tree T a dendrex in $\mathrm{N}\boldsymbol{M}_T$ can be pictured as follows. It consist of an assignments c of colors of $\boldsymbol{M}$ to the edges of T and a equivalence class of pairs (p, f), where p is a planar structure on T and f assigns an operation $f(v) \in \boldsymbol{M}(c(e_1), \ldots, c(e_n); c(e))$ to each vertex $v \in T$, where e denotes the output edge of v and $e_1, \ldots, e_n$ enumerates the input edges of v in the order provided by the planar structure p. Two such pairs (p, f) and (q, g) are equivalent if for each vertex v, the permutation σ relating the planar structure p and q satisfies $f(v) = \sigma^* g(v)$.

A little more informally, a dendrex in $\mathrm{N}\boldsymbol{M}_T$ is an assignment of colors of $\boldsymbol{M}$ to the edges of T and a compatible assignment of an operation in $\boldsymbol{M}$ to each vertex of T. As explained above, this is only a precise statement if one has a planar structure on T in mind.

Example 2.5.6. Every simplicial set S gives a dendroidal set $i_! S$ by 'extension by zero',

$$(i_! S)_T = \begin{cases} S_n & \text{if } T \simeq [n] \\ \varnothing & \text{otherwise} \end{cases}$$

As the notation suggest, the functor

$$i_! : \mathsf{s}\mathcal{S}\mathsf{et} \to \mathsf{d}\mathcal{S}\mathsf{et}$$

is the left Kan extension of functor

$$i : \mathbf{\Delta} \to \mathsf{d}\mathcal{S}\mathsf{et}$$

which sends $[n]$ to the representable $\Omega[n]$. Its right adjoint $i^* : \mathsf{d}\mathcal{S}\mathsf{et} \to \mathsf{s}\mathcal{S}\mathsf{et}$ gives for every dendroidal set D its underlying simplicial set $i^* D$.

Following suit with simplicial sets, we can define the notion of degenerate and non-degenerate dendrices. We will use them later to give a more explicit description of the left adjoint $h : \mathsf{d}\mathcal{S}\mathsf{et} \to \mathsf{Op}$ to the nerve functor $\mathrm{N} : \mathsf{Op} \to \mathsf{d}\mathcal{S}\mathsf{et}$.

Definition 2.5.7. Let D be a dendroidal set. A dendrex $d \in D_T$ is called *degenerate* if there is a degeneracy map $\sigma : T \to S$ which is not an isomorphism and a dendrex $y \in X_S$ with $\sigma^* y = d$. A dendrex is *non-degenerate* if such a pair does not exist.

2.6 Connections to other Categories

We may use the fact that $\mathsf{d}\mathcal{S}\mathsf{et}$ is a presheaf category to construct adjunctions between it and other categories using Kan extensions. We care only about two: adjunctions that connect dendroidal sets to simplicial sets and adjunctions that connect dendroidal sets to operads. These constructions reinterpret examples from the previous sections as functors.

Remark 2.6.1 (Dendroidal Sets and Operads). Recall from Example 2.5.5, that every operad in $\mathcal{S}\mathrm{et}$ has a *nerve* $\mathrm{N}\boldsymbol{P}$. This construction defines an adjoint pair

$$\begin{tikzcd} \mathsf{d}\mathcal{S}\mathsf{et} \arrow[r, shift left, "h"] \arrow[r, phantom, "\perp"] & \mathsf{Op} \arrow[l, shift left, "\mathrm{N}"] \end{tikzcd}$$

Up to natural isomorphism, the left adjoint h is the unique colimit-preserving functor which coincides with $\boldsymbol{\Omega}$ on representables, i.e.

$$h(\boldsymbol{\Omega}[T]) = \boldsymbol{\Omega}[T]$$

More explicitly, for a dendroidal set X, the operad $h(X)$ has as its colors the elements of the set X_μ, while the operations are generated under composition by elements of X_{C_n} for $n \geq 0$. These are subject to the following two types of relations: a degenerate element of X_{C_1} corresponding to a color $c \in X_\mu$ gives the identity morphism id_c, whereas the operation corresponding to an inner face $\partial_e T$ of a tree T with two vertices is identified with the composition of the two operations corresponding to the vertices f and g of T.

There is a simplified description of $h(X)$ for dendroidal sets satisfying the *inner Kan condition*, but this will be discussed further in **BACK LINK TO FUTURE CONSTRUCTION NEEDED HERE**. The nerve functor N is fully faithful. **CITATION NEEDED**

Remark 2.6.2 (Dendroidal Sets and Simplicial Sets). The inclusion $i : \boldsymbol{\Delta} \to \boldsymbol{\Omega}$

defines three adjoint functors

$$\begin{tikzcd} \mathsf{s}\mathcal{S}\mathsf{et} \arrow[rr, bend left, "i_!"] \arrow[rr, bend right, "i_*"'] & {} & \mathsf{d}\mathcal{S}\mathsf{et}. \arrow[ll, "i^*" description] \end{tikzcd}$$

Thus, each dendroidal set X has an underlying simplicial set i^*X defined by

$$(i^*X)_n = X_{[n]}.$$

Conversely, a simplicial set Y gives rise to a dendroidal set $i_!Y$ by 'extension by zero', as already mentioned in Example 2.5.6. Since $i : \mathbf{\Delta} \to \mathbf{\Omega}$ is fully faithful, so are both $i_!$ and i^*. Thus, we can identify the category of simplicial sets with a full subcategory of the category of dendroidal sets. More precisely, we have the following equivalence:

$$\mathbf{\Delta} \simeq \mathbf{\Omega}/\mu.$$

We will prove that this actually implies that

$$\mathsf{s}\mathcal{S}\mathsf{et} \simeq \mathsf{d}\mathcal{S}\mathsf{et}/\mathbf{\Omega}[\mu].$$

To prove the previous claim we will use the following lemma

Lemma 2.6.3. *Let* C *be any small category and* $X \in \mathcal{S}\mathsf{et}^{\mathsf{C}^{\mathrm{op}}}$. *Then there is an equivalence of categories*

$$\mathcal{S}\mathsf{et}^{\mathsf{C}^{\mathrm{op}}}/X \simeq \mathcal{S}\mathsf{et}^{(\int X)^{\mathrm{op}}}$$

Proof. We will construct an equivalence between the two categories. Let $\alpha : Z \to X$ be an object of $\mathcal{S}\mathsf{et}^{\mathsf{C}^{\mathrm{op}}}/X$. We define a functor $F : (\int X)^{\mathrm{op}} \to \mathcal{S}\mathsf{et}$ as follows. For an object (c,x) of $\int X$, we set

$$F(c,x) = \{z \in Z_c \mid \alpha_c(z) = x\},$$

i.e. the preimage of x under the leg α_c of the natural transformation. We denote this set by $\alpha_c^{-1}(x)$. For a morphism $(c,x) \xrightarrow{g} (d,y)$ in $\int X$, we define $Fg : \alpha_d^{-1}(y) \to \alpha_c^{-1}(x)$ by sending $w \in \alpha_d^{-1}(y)$ to $Zg(w)$. This is well defined since by the naturality of α and by the definition of morphism in $\int X$, we have $\alpha_c(Zg(w)) = Xg(\alpha_d(w)) = Xg(y) = x$. Thus, we have defined a functor $F : (\int X)^{\mathrm{op}} \to \mathcal{S}\mathsf{et}$.

Conversely, let $G : (\int X)^{\mathrm{op}} \to \mathcal{S}\mathsf{et}$ be an object of $\mathcal{S}\mathsf{et}^{(\int X)^{\mathrm{op}}}$. We define a natural transformation $\gamma : Z \to X$ as follows. For an object c of C, we set $Z_c = \coprod_{x \in X_c} G(c,x)$ and define $\gamma_c : Z_c \to X_c$ by sending an element $z \in G(c,x)$ to x. For a morphism $c \xrightarrow{f} d$ in C, we define $Zf : Z_d \to Z_c$ by sending an element $z \in G(d,y)$ to the element $Gf(z) \in G(d, Xf(x)) \subseteq Z_c$. This is well defined since for any element $z \in G(c,x)$, we have **Need to finish the final touches plus show that is a equivalence** $\square$

These two connection give rise to the following commutative square of functors where the square of left adjoint commutes up to natural isomorphism, as does the square of right adjoints:

$$\begin{tikzcd}
\mathsf{s}\mathcal{S}\mathsf{et} \arrow[r, shift left, "h"] \arrow[d, shift right, "i_!"'] & \mathcal{C}\mathsf{at} \arrow[l, shift left, "\mathrm{N}"] \arrow[d, shift right, "j_!"'] \\
\mathsf{d}\mathcal{S}\mathsf{et} \arrow[r, shift left, "h"] \arrow[u, shift right, "i^*"'] & \mathsf{Op}. \arrow[l, shift left, "\mathrm{N}"] \arrow[u, shift right, "j^*"']
\end{tikzcd}$$

The commutativity of the squares translates into natural isomorphism of functors $j_! h \simeq h i_!$ and $\mathrm{N} j^* \simeq i^* \mathrm{N}$. Note that since $i_!$, $j_!$ and both nerve functors are fully faithful, we also have the following natural isomorphism $\mathrm{N} j_! \simeq i_! \mathrm{N}$ and $h i^* \simeq j^* h$. **PUT LEMMA HERE ABOUT THIS**

2.7 Tensor Product in $\mathsf{d}\mathcal{S}\mathsf{et}$

Recall from 2.1.1 that the category of operads Op has a monoidal structure given by the Boardman-Vogt tensor product. Since the nerve functor $\mathrm{N} : \mathsf{Op} \to \mathsf{d}\mathcal{S}\mathsf{et}$ is fully faithful, we can transport this monoidal structure to $\mathsf{d}\mathcal{S}\mathsf{et}$ and obtain a monoidal structure on it. We will do this by by first define a functor

$$ten : \mathbf{\Omega} \times \mathbf{\Omega} \to \mathsf{d}\mathcal{S}\mathsf{et}$$

by assigning to a pair of trees (S, T) the corresponding free operads $\Omega(S)$ and $\Omega(T)$, forming their tensor product $\Omega(S) \otimes_{BV} \Omega(T)$, and finally taking the dendroidal nerve of the resulting operad:

$$ten(S, T) = \mathrm{N}(\Omega(S) \otimes_{BV} \Omega(T)).$$

By the method of Kan extension, this functor determines a functor

$$\otimes : \mathsf{d}\mathcal{S}\mathsf{et} \times \mathsf{d}\mathcal{S}\mathsf{et} \to \mathsf{d}\mathcal{S}\mathsf{et}$$

uniquely determined up to natural isomorphism by the requirement that it preserves colimits in each variable and that it coincides with *ten* on representables, i.e.

$$\Omega(S) \otimes \Omega(T) = \mathrm{N}(\Omega(S) \otimes_{BV} \Omega(T)),$$

$$\varinjlim_i (X \otimes Y_i) \simeq X \otimes \varinjlim_i Y_i,$$

$$\varinjlim_i (X_i \otimes Y) \simeq \varinjlim_i X_i \otimes Y.$$

The category $\mathsf{d}\mathcal{S}\mathsf{et}$, being a presheaf category, is Cartesian closed and therefore has inner hom objects adjoint to the cartesian product. However, these inner homs do not model the correct type of morphisms that we want for the category of dendroidal sets to be a combinatorial model for the homotopy theory of operads. Instead, we need an inner hom object that adjoint to the tensor product we have just defined.

Definition 2.7.1. The inner hom object which is right adjoint to the tensor product is denoted by **Hom**. This internal hom is characterized by natural isomorphism

$$\mathsf{d}\mathcal{S}\mathsf{et}(X \otimes Y, Z) \simeq \mathsf{d}\mathcal{S}\mathsf{et}(X, \mathbf{Hom}(Y, Z))$$

for dendroidal sets X, Y, Z.

We will use extensively this object and the underlying simplicial set of these hom objects.

$$\mathbf{hom}(Y, Z) := i^* \mathbf{Hom}(Y, Z).$$

Proposition 2.7.2. *The tensor product of dendroidal sets satisfies the following* *properties:*

(i) It is symmetric

(ii) The functor $h : \mathsf{d}\mathcal{S}\mathsf{et} \to \mathsf{Op}$ commutes with tensor products, i.e. there is a natural isomorphism of operads $h(X \otimes Y) \simeq h(X) \otimes_{BV} h(Y)$.

(iii) The functor $i_! : \mathsf{s}\mathcal{S}\mathsf{et} \to \mathsf{d}\mathcal{S}\mathsf{et}$ send product to tensor products, i.e. there is a natural isomorphism $i_!(M \times N) \simeq i_!(X) \otimes i_!(Y)$ for simplicial sets M, N.

Proof. Since the functors $\otimes$, h, and $i_!$ are all compatible with colimits it suffices to establish these natural isomorphisms for representables. Then (i) follows from the symmetry of the Boardman-Vogt tensor product. For (ii), recall from Remark 2.6.1 that the nerve functor N is fully faithful, so that the counit $\epsilon : h\mathrm{N} \to \mathrm{id}$ is a natural isomorphism. It follows that

$$h(\Omega[S] \otimes \Omega[T]) = h\mathrm{N}(\Omega(S) \otimes_{BV} \Omega(T))$$

$$\simeq \Omega(S) \otimes_{BV} \Omega(T)$$

$$h(\Omega[S] \otimes \Omega[T]) = h(\mathrm{N}\Omega(S)) \otimes_{BV} h(\mathrm{N}\Omega(T))$$

Finally, (iii) follows from the fact that for categories C and D the tensor product $\mathsf{C} \otimes_{BV} \mathsf{D}$ coincides with the cartesian product $\mathsf{C} \times \mathsf{D}$, so that for simplicial sets

M, N we have

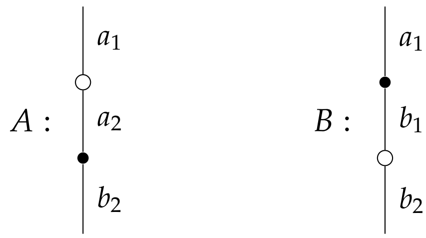
$$
\begin{aligned}
i_!(M \times N)_{[n]} &= (M \times N)_n \\
&\simeq M_n \times N_n \\
&\simeq i_! M_{[n]} \times i_! N_{[n]} \\
&\simeq (i_! M \otimes i_! N)_{[n]}
\end{aligned}
$$

□

Remark 2.7.3. We will note that the tensor product of dendroidal sets is not associative up to coherent isomorphism. This is the main issue at hand we will need to address in the rest of the thesis. In good news, we will see that the tensor product is going to be associative up to homotopy, and that this is good enough for our purposes.

Example 2.7.4. This example is meant to illustrate Proposition 2.7.2 (iii). Given the simplicial set $[1]$, we can compute the tensor product $i_![1] \otimes i_![1]$ as follows. First, we can see that $i_![1]$ is given by the representable $\mathbf{\Omega}[1]$, given by the linear tree:

To simplify notation we write $a_1 := a \otimes_{BV} 1$, $a_2 := a \otimes_{BV} 2$, $b_1 := b \otimes_{BV} 1$, and $b_2 := b \otimes_{BV} 2$ for the objects in the Boardman-Vogt tensor product. Then, we can compute the Boardman-Vogt tensor product by drawing the shuffle trees, which are given by the following trees:



Where we note that the Boardman-Vogt interchange relation identifies the full composition of the full tree A with the full composition of the full tree B. Since the shuffle trees are also linear, we can think of them as simplices in the following way:

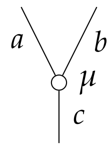
$$A: \begin{tikzcd} & a_2 \arrow[dr] \arrow[d, phantom, "\shortparallel"] & \\ a_1 \arrow[ur] \arrow[rr] & {} & b_2 \end{tikzcd} \qquad B: \begin{tikzcd} & b_1 \arrow[dr] \arrow[d, phantom, "\shortparallel"] & \\ a_1 \arrow[ur] \arrow[rr] & {} & b_2 \end{tikzcd}$$

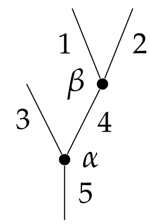
And what the Boardman-Vogt interchange relation says that we have an equality between the composite of A and B, that is:

$$\begin{tikzcd} a_1 \arrow[r] \arrow[d] \arrow[dr, bend left] \arrow[dr, bend right] \arrow[dr, phantom, "="] & b_1 \arrow[d] \\ a_2 \arrow[r] & b_2 \end{tikzcd}$$

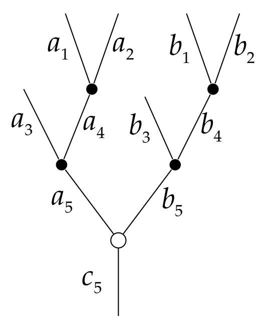
Which is precisely what we were expecting to get from the fact that $i_![1] \otimes i_![1]$ is isomorphic to $i_!([1] \times [1])$.

Example 2.7.5. Let us examine the tensor product of two small representable dendroidal sets $\Omega[S]$ and $\Omega[T]$, letting the trees S and T be as below

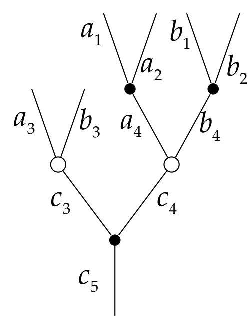
S : 

T : 

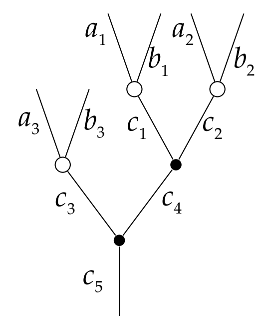
By definition, we can compute the Boardman-Vogt tensor product of the free operads $\Omega(S)$ and $\Omega(T)$, and then take the nerve of the resulting operad. The colors of $\Omega(S) \otimes_{BV} \Omega(T)$ are given by the cartesian product of the colors of $\Omega(S)$ and $\Omega(T)$, which are in turn given by the edges of S and T. The operations of $\Omega(S) \otimes_{BV} \Omega(T)$ are generated under composition by the operations of $\Omega(S)$ and $\Omega(T)$, which are given by the vertices of S and T. We can describe all operations by subtrees of the following trees, called shuffle trees:



A



B



C

Here is one one intuition for shuffle trees, we can see we can get tree B from A by *shuffling up* the white vertex by its input edges. We can see that the representation of an operation in $\Omega(S) \otimes_{BV} \Omega(T)$ as a subtree of A, B or C might be not unique, but given the Boardman-Vogt Interchange relation these representations will all be identified with each other. What this all means for the dendroidal set given by taking the nerve is the following. The dendroidal set is given by the union of the representables given by the shuffle trees, glued together by the Boardman-Vogt interchange relation. In particular, we will have the following diagram in the category of Ω :

$$
\begin{tikzcd}
 & \partial_{a_5}\partial_{b_5}A \arrow[dr] \arrow[dd, "\cong"'] & \\
C_A \arrow[ur] \arrow[dd, "\cong"] & & A \\
 & \partial_{c_3}\partial_{c_4}B \arrow[dr] & \\
C_B \arrow[ur] \arrow[dr] \arrow[dd, "\cong"] & & B \\
 & \partial_{a_4}\partial_{b_4}B \arrow[ur] \arrow[dd, "\cong"] & \\
C_C \arrow[dr] & & C \\
 & \partial_{c_1}\partial_{c_2}C \arrow[ur] &
\end{tikzcd}
$$

In this diagram, $C_A := \partial_{b_4}\partial_{a_4}\partial_{b_5}\partial_{a_5}A$, C_B, and C_C are the corollas given by contracting all the inner edges of A, B, and C respectively. All of these corollas are identified with each other by the Boardman-Vogt interchange relation, and thus they are isomorphic. For the objects in the middle, we can observe that A and B hace a common face given by contracting the edges a_5, b_5 for A and c_3, c_4 for B. Since the corollas are faces of these two objects, they have a map from the corollas to this intermidiate faces. The bottom part of the diagram is described similarly.

Then, we can compute the dendroidal set $\Omega[S] \otimes \Omega[T]$ by the colimit of this diagram, viewed as a diagram of representables.

Example 2.7.6. **ADD EXAMPLE TO SHOW FAILURE OF ASSOCIATIVITY**

2.8 Normal dendroidal sets and monomorphism

We will now introduce the notion of normal dendroidal sets, which are the dendroidal sets that are cofibrant in the operadic model structure. We will see that these are precisely the dendroidal sets that can be built up from representables by attaching cells along monomorphisms. We will also see that the tensor product of normal dendroidal sets is again normal, and that the tensor product of monomorphisms is again a monomorphism. We can compare the role of the normal dendroidal sets in the operadic model structure to the role of CW complexes in the homotopy theory of topological spaces.

Definition 2.8.1. A dendroidal set X is called *normal* if for every tree T, the action of the automorphism group $\mathrm{Aut}(T)$ on the set of T-dendrices X_T is free.

More generally we can define the notion of normal monomorphisms

Definition 2.8.2. Given a monomorphism of dendroidal sets $i : X \to Y$, we say that i is a *normal* if for every tree T, the action of $\mathrm{Aut}(T)$ on the complement of the image, $Y_T - i(X_T)$, is free.

Lemma 2.8.3. *Every representable dendroidal sets are normal.*

Proof. Assume that $f : S \to T$ is a morphism of trees, and $\alpha \in \mathrm{Aut}(S)$ such that $f\alpha = f$. You can factor f as $\delta\beta\sigma$, where σ is a degeneracy, β is an isomorphism, and δ is a face map, as mentioned in Remark 2.4.9. Since $\delta\beta$ is a monomorphism, we must have that $\sigma\alpha = \sigma$. Since σ is a degeneracy, σ is an epimorphim, which implies that α must be the identity. □

Lemma 2.8.4. *Let P be an operad with the property that for each object x, the symmetric group Σ_n acts freely on the set*

$$P_n(-;x) := \coprod_{x_1,\ldots,x_n} P(x_1,\ldots,x_n;x)$$

for all n-ary operations into x. Then the dendroidal nerve $\mathrm{N}P$ is a normal dendroidal set.

Proof. To see this, note that by assumption $\mathrm{Aut}(C_n)$ acts freely on $\mathrm{N}P_{C_n}$ for each $n \geq 0$. Now we will proceed by induction on the size of the trees. Given a tree T, $\alpha \in \mathrm{Aut}(T)$ such that $\alpha^*\xi = \xi$ for $\xi \in \mathrm{N}P_T$. If we denote the root vertex of T as v_r and the subtree given by the vertex f such that $O(f) = v_r$ by C_{v_r} with inputs $e_1, \ldots, e_n$, with natural inclusion $j : C_{v_r} \to T$. We can see that α restricts to $\alpha_r \in \mathrm{Aut}(C_{v_r})$, and thus $\alpha_r^*(j^*\xi) = j^*\xi$, so that α_r must be the identity. It follows that α restricts to an automorphism α_i of each of the subtrees T/e_i given by removing the vertex f and its output, and taking the subtbree where e_i is the root. The restriction of ξ to T/e_i is fixed by α_i, and thus by induction, α_i must be the identity for each i. It follows that α must be the identity, and thus $\mathrm{N}P$ is normal. □

We will use the following result but do not prove it here. If interested, the proof can be found in Moerdijk and Weiss, 2007 originally or in Heuts and Moerdijk, 2022 proposition 3.29. The proof consist of using Quillen's small object argument to factor every morphism into a normal monomorphism followed by a *trivial fibration*, cf definition 2.8.8.

Definition 2.8.5. Every morphism of dendroidal sets can be factored as a normal monomorphism followed by a morphism that has the right lifting property with respect to all normal monomorphisms.

Proposition 2.8.6. *(i) If $f : A \to B$ and $g : B \to C$ are normal monomorphisms, then $g \circ f$ is a normal monomorphism.*

(ii) If

$$\begin{tikzcd} A \arrow[r] \arrow[d, "g^*"'] \arrow[dr, phantom, very near start, "\lrcorner"] & C \arrow[d, hook, "g"] \\ B \arrow[r] & D \end{tikzcd}$$

is a pullback square where g is normal, then g^* is a normal monomorphism.

(iii) If $f : A \to B$ is a morphism of dendroidal sets and B is normal, the A is normal

(iv) If

$$\begin{tikzcd} A \arrow[r] \arrow[d, hook, "f"'] & C \arrow[d, "f^*"] \\ B \arrow[r] \arrow[ur, phantom, very near end, "\ulcorner"] & D \end{tikzcd}$$

is a pushout square where f is a normal monomorphism, then f^* is a normal monomorphism.

(v) If $u : A \to B$ is a retract of $v : C \to D$, then u is a normal monomorphism if v

is a normal monomorphism.

(vi) If $\{f_i : A_i \to B_i\}_{i \in I}$ is a famlily of normal monomorphisms, then their coproduct

$$\coprod_{i \in I} f_i : \coprod_{i \in I} A_i \to \coprod_{i \in I} B_i$$

is a normal monomorphism.

(vii) If $A_0 \to A_1 \to A_2 \to \cdots$ is a sequence with colimit A_∞ for which each $f_i : A_i \to A_{i+1}$ is a normal monomorphism, then each cone leg morphism $A_i \to A_\infty$ is a normal monomorphism.

The previous proposition allows us to give the following characterixation of normal monomorphism

Theorem 2.8.7. *The class of normal monomorphisms is the smallest weakly saturated class containing the boundary inclusions $\partial\Omega[T] \to \Omega[T]$ for every tree $T \in \Omega$*

Definition 2.8.8. We will call a morphism of dendroidal sets a *trivial fibration* if it has the right lifting property with respect to all normal monomorphisms. That is given a commutative square as below, where i is a normal monomorphism and p is a trivial fibration, there exists a diagonal filler h making the whole diagram commute.

$$\begin{tikzcd} A \arrow[r] \arrow[d, tail, "i"'] & B \arrow[d, "p"] \\ C \arrow[r] \arrow[ur, dashed, "h"] & D \end{tikzcd}$$

Before moving on to the next section, we will show some important examples of normal monomorphisms, which will be used throughout the rest of the thesis. Specifically, we want to undertand how normal monomorphisms interact with the tensor product. We will do this by fixing a (normal) dendroidal set X and tensoring with objects with normal monomorphisms between them. We also want to check if the *pushout-product* of a normal monomorphisms and a monomorphism of simplicial sets is a normal monomorphisms. We will not prove these results here, but they can be found in Heuts and Moerdijk, 2022 proposition 4.21 and forward.

We start with the pushout-product proposition. This will help us later show that dendroidal sets are "enriched" compatibly with the Joyal model structure of simplicial sets.

Proposition 2.8.9. *Let $f : M \to N$ be a monomorphism of simplicial sets, and $g : X \to Y$ be a normal monomorphism of dendroidal sets. Then the pushout-product*

$$i_!M \otimes Y \coprod_{i_!M \otimes X} i_!N \otimes X \to i_!N \otimes Y$$

is again a normal monomorphism.

For the study of the tensor product, we need to define an unbiased way to tensor several objects.

Definition 2.8.10. We can define the n-fold tensor product of dendroidal sets as a functor

$$\otimes_n : \mathsf{d}\mathcal{S}\mathsf{et}^{\times n} \to \mathsf{d}\mathcal{S}\mathsf{et}$$

which is unique up to isomorphism having the requirement that:

1. It preserves colimits in each of its n variables separetely.

2. $\otimes_n(T[\Omega_1], \ldots, T[\Omega_n]) = \mathrm{N}(T(\Omega_1), \ldots, T(\Omega_n))$

We will write $\otimes_n(X_1, \dots, X_n) = X_1 \otimes \dots X_n$ for general dendroidal sets $X_1, \dots, X_n$. As mentioned before, the tensor product on dendroidal sets is not associative up to isomorphism, but associator maps, even canonical ones, exist for it. We will denote them, for $0 \leq i < j \leq n$ as

$$\alpha_{i,j} : X_1 \otimes \dots X_{i-1} \otimes (X_i \otimes \dots X_j) \otimes X_{j+1} \otimes \dots X_n \to X_1 \otimes \dots X_n$$

where we may skip the indexing if clear by context. This are natural in each of the n variables.

Now we can talk about the normality properties of this associators. For the following proposition we will conflate the notation of tree R, S, and T with the representable in dendroidal sets.

Proposition 2.8.11. *Each associator*

$$\alpha : R_1 \otimes \dots R_k \otimes (S_1 \otimes \dots S_l) \otimes T_1 \otimes \dots T_m \to R_1 \otimes \dots \otimes T_m$$

is a normal monomorphism, for trees $R_1, \dots, R_k$, etc.

Corollary 2.8.12. *The associativity map α in the previous proposition is an isomorphism whenever $R_1, \dots, R_k$ and $T_1, \dots, T_m$ are all linear trees, i.e. simplices.*

Corollary 2.8.13. *Let $X_1, \dots, X_n$ be normal dendroidal sets. Then each associativity map*

$$\alpha_{i,j} : X_1 \otimes \dots X_{i-1} \otimes (X_i \otimes \dots X_j) \otimes X_{j+1} \otimes \dots X_n \to X_1 \otimes \dots X_n$$

is a normal monomorphism. It is an isomorphism if $X_1, \dots, X_{i-1}$ and $X_{j+1} \otimes \dots \otimes X_n$ are simplicial sets (viewed as dendroidal sets via the embedding $i_!$)

Bibliography

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Moerdijk, I. and I. Weiss (2007). “Dendroidal sets”. In: *Algebr. Geom. Topol. 7*.